SET A

Register No	R	A	2	5	1	2	0	4	4	0	1	5	0	7	1

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
21CSC555J – MACHINE LEARNING ALGORITHMS
CYCLE TEST – I
Theory

Program Offered: M. Tech
Max. Marks: 40

Year / Semester: I/ I
Duration: 1.5 Hrs
Date of Exam: 27/09/25

PART A (2 x 20 =40 Marks)

Answer any Two Questions

1 Explain how cross-validation, particularly K-fold cross-validation, helps in improving model performance and reduces the chances of overfitting. Provide examples of how it compares to using a simple train-test split. Draw K-fold CV flow diagram. **(20 Marks)**

2 You have built a binary classifier to predict whether emails are spam or not spam. After testing your classifier on a dataset of 500 emails, you obtain the following confusion matrix

N=500	Actual Not Spam	Actual Spam
Predicted Not Spam	350	20
Predicted Spam	30	100

a) Calculate the overall accuracy of the classifier **(5 Marks)**
b) Calculate the precision, recall, and F1-score for each class. **(10 Marks)**
c) Analyze the classifier's performance based on these metrics and discuss which class the model struggles with the most. **(5 Marks)**

3 i. Explain Linear regression with suitable examples. **(10 Marks)**

ii. Explain in detail about logistic regression classifier, advantages and applications with suitable example? **(10 Marks)**